

SENIOR PROGRAMMER

Beau Harrison

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Summary

- Leading the modernization of legacy control systems in adopting new technology and integrating industry best practices.
- Explored text analysis and data visualization as a part of the University of Chicago's Public Policy master's program.
- Built web applications to streamline particle accelerator operations at the United States' largest physics laboratory.
- Developed an educational curriculum to bring professional tools to middle school students while analyzing image data.

Select Skills (Needs work)

- Ruby on Rails (learning)
- Typescript/JavaScript
- HTML/CSS
- Java/C++
- Linux
- SQL (Postgres/MariaDB/SQLite)
- Git
- GitHub Actions
- Ansible
- GraphQL/gRPC

Experience Highlights

Software Engineering Manager - 2022 to present

- Building two software development teams to modernize ~100 accelerator applications - 2025 to present
- Establishing software development plan and align with industry - 2024 to present
- Establish architecture and design for modern control system - 2022 to present

Application Developer & System Analyst - 2018 to 2022

- Leading 5G mmWave feasibility study for underground accelerator enclosures - 2021 to 2023
- Establishing use-cases for a 10 year DOE O413.3B control system modernization effort - 2020 to present
- Championed saving in overhead and 20 FTEs through DevOps upgrades - 2018 to 2020
- Leading technicians while architecting a new personnel protection system for particle accelerators - 2018 to 2022
- Managing the design of APIs across languages, balancing stakeholder priorities - 2016 to present

Senior Accelerator Operator - 2015 to 2018

- Managed the DOE Proton Improvement Plan management board in \$10M accelerator upgrades- 2015 to 2018
- Created a 90% cost-reducing flexible hardware interface replacement - 2017
- Served as a consultant in response to accelerator access safety violation - 2016
- Led a team to convert legacy documentation platform to adopt industry-standard - 2015 to 2018

Accelerator Operator II - 2012 to 2015

- Developed and maintaining multiple essential web applications for core accelerator operation - 2013 to present
- Authored particle accelerator concepts reference utilized by international accelerator community - 2012 to 2013
- Facilitated and coordinated accelerator operations tasks while serving as interim Crew Chief - 2012 to 2018

Accelerator Operator I - 2011 to 2012

- Tuned and operated the accelerator facility - 2011 to 2018
- Trained on Fermi National Accelerator Laboratory accelerator systems - 2011 to 2012

Education

University of Chicago (2014-2018) Chicago, IL
Master of Public Policy

Henderson State University (2007-10) Arkadelphia, Arkansas
Bachelor of Science Degree in Physics